

Pokémon X & Y Base Stat Analysis



Pratheek Pala

San José State University

December 9, 2024

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Introduction

The stats of Pokémon in the Pokémon games are very diverse and intriguing. It includes things such as each Pokémon's attack and defense, special attack and defense, whether the Pokémon is legendary or not, and even the total of all its stats. Upon observing the data, questions were formulated on points of interest:

1. What would be the total base stat for an average Pokémon?

In asking this question about the data set, we are essentially asking how strong the average Pokémon is. Through analyzing the data, we hope to answer the question and discern which method is the best measure of this value.

2. Do legendary Pokémon have statistically higher mean total base stats than all other Pokémon?

What we want to know through answering this question is if legendary Pokémon are truly special just based off of stats or if they are simply called legendary because of their rarity.

3. Are the sums of special attack and special defense related to the sums of regular attack and regular defense?

By answering this question we want to see if a Pokémon with stronger attack and defense also has stronger special attack and defense.

Data

	#	Name	Type 1	Type 2	Total	HP	Attack	Defense	Sp. Atk	Sp. Def	Speed	Generation	Legendary
0	1	Bulbasaur	Grass	Poison	318	45	49	49	65	65	45	1	False
1	2	Ivysaur	Grass	Poison	405	60	62	63	80	80	60	1	False
2	3	Venusaur	Grass	Poison	525	80	82	83	100	100	80	1	False
3	3	VenusaurMega Venusaur	Grass	Poison	625	80	100	123	122	120	80	1	False
4	4	Charmander	Fire	NaN	309	39	52	43	60	50	65	1	False
...	...	...	...	...	...	...	...	...	...	...	...	...	...
795	719	Diancie	Rock	Fairy	600	50	100	150	100	150	50	6	True
796	719	DiancieMega Diancie	Rock	Fairy	700	50	160	110	160	110	110	6	True
797	720	HoopaaHoopaa Confined	Psychic	Ghost	600	80	110	60	150	130	70	6	True
798	720	HoopaaHoopaa Unbound	Psychic	Dark	680	80	160	60	170	130	80	6	True
799	721	Volcanion	Fire	Water	600	80	110	120	130	90	70	6	True

One of the columns analysis was conducted on was the ‘Total’ column. This column is the sum of HP, attack, defense, special attack, special defense, and speed. The higher the number in each category, the better the Pokémon is in that category. A higher total stat typically means a stronger Pokémon.

Some other columns analyzed include the ‘Attack’ and ‘Defense’ columns, and the ‘Sp. Attack’ and ‘Sp. Defense’ columns. Pokémon in the Pokémon games have normal attacks as well as special attacks. The stats in the columns above measure a Pokémon's ability to execute and defend against each respective attack.

Another column that was observed was the ‘Legendary’ column. This column contains boolean values that tell us if a Pokémon is legendary or not.

None of the columns selected have missing values, and they all have a length of 800.

Methods

Some methods used to conduct statistical analysis include data visualization, central tendency measures, correlation, and a z-test (hypothesis test).

- Data visualization includes a histogram to visualize the distribution of 'Total,' and a scatterplot to observe the relationship between the sums of 'Attack' and 'Defense' and of 'Sp. Attack' and 'Sp. Defense.' A histogram was used because knowing the distribution of the 'Total' data is helpful when conducting a hypothesis test and deciding upon a measure of central tendency. A scatterplot was used in order to determine if Pearson correlation could be used. In order to clearly see the relationship between both variables, outliers were removed in the datasets of both variables.

```
[ ]
sp_sorted = poke_df['Sp. Sum'].sort_values().to_frame()
sp_sorted.columns = ['Sp. Sum']

reg_sorted = poke_df['Reg Sum'].sort_values().to_frame()
reg_sorted.columns = ['Reg Sum']

sq1 = np.percentile(sp_sorted, 25)
sq3 = np.percentile(sp_sorted, 75)
siqr = sq3 - sq1

rq1 = np.percentile(reg_sorted, 25)
rq3 = np.percentile(reg_sorted, 75)
riqr = rq3 - rq1

new_sp = sp_sorted.loc[(sp_sorted['Sp. Sum'] >= (sq1 - 1.5*siqr)) & (sp_sorted['Sp. Sum'] <= (sq3+1.5*siqr))]

new_reg = reg_sorted.loc[(reg_sorted['Reg Sum'] >= (rq1 - 1.5*riqr)) & (reg_sorted['Reg Sum'] <= (rq3+1.5*riqr))]

sums_merged = pd.concat([new_sp, new_reg], axis=1)

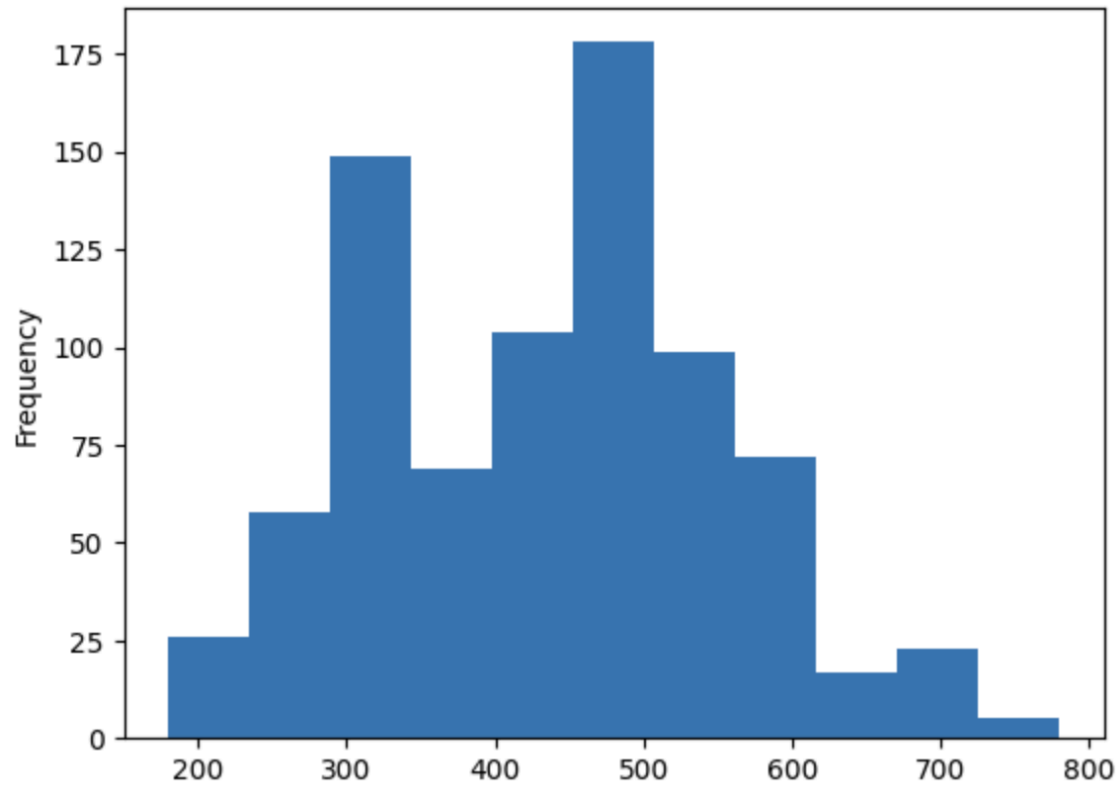
sns.relplot(data = sums_merged, x = 'Sp. Sum', y = 'Reg Sum').set(title = "Relationship Between Special Sum and Regular Sum")
```

- The central tendency measure used was median. This was used because the data was not normally distributed. Though there were no outliers in the 'Total' column, median is still the best choice as it is the most accurate when dealing with a non-normal distribution.

- Pearson correlation was used when finding the relationship between the sums of 'Attack' and 'Defense' and of 'Sp. Attack' and 'Sp. Defense' to determine if there is a relationship between them. Correlation was used to determine if there was a relationship since the data is linear and the range of potential values in the data is large as shown by the scatterplot.
- Z-test was used to assess if legendary Pokémon have statistically higher total base stats than non-legendary Pokémon. A z-test was deemed the most appropriate since the population standard deviation is known, and the length of both samples is large ($n > 30$). The large length of the samples is used as justification by the Central Limit Theorem, which states that the sampling distribution of sample means approaches a normal distribution as n increases, regardless of whether the population distribution is normal or not.

Results

Visual Representation A:



This histogram tells us that the distribution of the data 'Total' is not normally distributed.

Results to Questions:

1. The total base stat for an average Pokémon would be 450, which was obtained by finding the median of the data set 'Total.'

```
import numpy as np
# find the number of outliers in total
total_sorted = poke_df['Total'].sort_values().to_frame()
total_sorted.columns = ['Total']

q1 = np.percentile(total_sorted, 25)
q3 = np.percentile(total_sorted, 75)
iqr = q3 - q1

new_total = total_sorted.loc[(total_sorted['Total'] >= (q1 - 1.5*iqr)) & (total_sorted['Total'] <= (q3 + 1.5*iqr))]
print(f'Number of outliers is {len(poke_df) - len(new_total)}')

# median
pokeMedian = poke_df['Total'].median()
print(f'Median: {round(pokeMedian,2)}')
```

Number of outliers is 0
Median: 450.0

2. The null hypothesis for this question was that legendary Pokémon do not have a statistically higher mean total base stat than all other Pokémon. The alternative hypothesis was that legendary Pokémon do have a statistically higher mean total base stat than all other Pokémon. A one-tail z-test was conducted because we only care about whether legendary Pokémon are statistically stronger, not just if their mean total base stat was statistically different from all other Pokémon. The test was conducted at an alpha level of 0.05, making the z-critical value about 1.65. Upon completing the test, it was found that the null hypothesis was rejected in favor of the alternative hypothesis. Thus,

legendary Pokémon do have a statistically higher mean total base stat than all other Pokémon.

```
# Creates one dataframe of just legendaries and one of all other pokemon
legendary = poke_df.loc[poke_df['Legendary'] == True]
all_others = poke_df.loc[poke_df['Legendary'] == False]

# finds the mean and length of both samples
legendary_mean = legendary['Total'].mean()
legendary_n = len(legendary)

all_others_mean = all_others['Total'].mean()
all_others_n = len(all_others)

# z score calculation
z = (legendary_mean - all_others_mean) / np.sqrt((pokeStd**2 / legendary_n) + (pokeStd**2 / all_others_n))
print(f"Z-Score: {round(z,2)}")

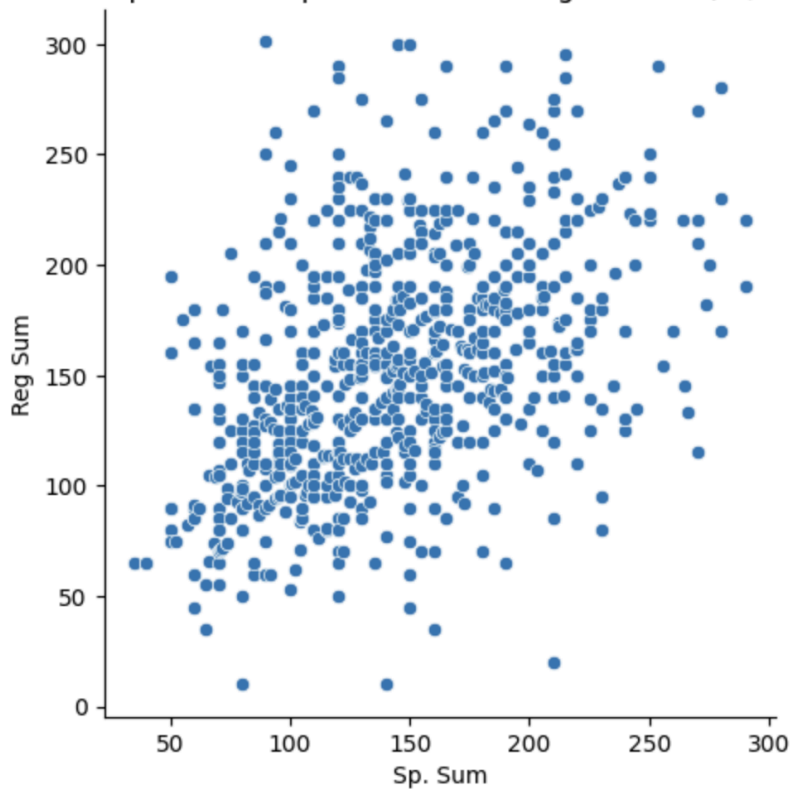
z_crit = 1.65 # at one tail, alpha = 0.05

if (z > z_crit):
    print("Null rejected.")
else:
    print("Null accepted.")
```

🔗 Z-Score: 14.19
Null rejected.

Visual representation B:

Relationship Between Special Sum and Regular Sum (w/o Outliers)



This scatterplot indicates there is a positive linear relationship between the sum of regular attack and defense and the sum of special attack and defense.

3. The null hypothesis for this question is that there is no relationship between the sums of special attack and special defense and the sums of regular attack and regular defense. The alternative hypothesis is that there is a relationship between the sums of special attack and special defense and the sums of regular attack and regular defense. Upon obtaining the Pearson correlation coefficient of about 0.47, we can understand that there is a fairly strong relationship between the two variables. The obtained p-value being much less than the alpha value of 0.05 tells us that the null hypothesis is rejected in favor of the

alternative. Thus, there is a relationship between the sums of special attack and special defense and the sums of regular attack and regular defense.

Discussion

The results found directly answer each of the questions about the data set, but statistical accuracy could be improved. A challenge faced during the analyses included misunderstanding the requirements of z-score standardization when dealing with a percent of a population above a certain value.

Conclusion

Some key takeaways include being careful when doing analysis to be sure that you are using an accurate method and making sure to document and handle any issues you may run into. Aside from this, analysis of the data has shown results that favor the formed hypotheses, which is cool to see.